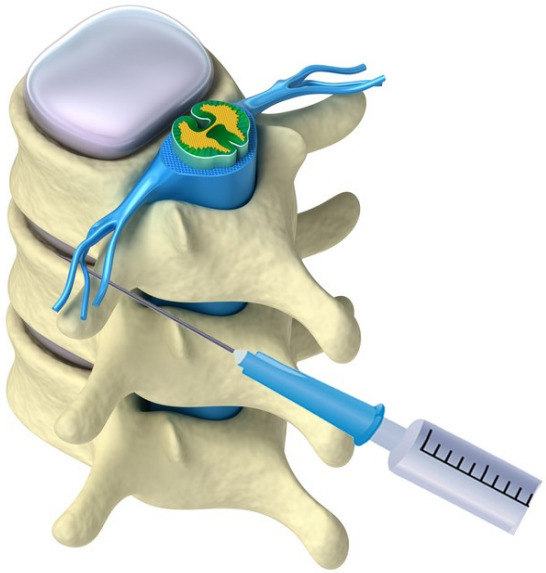
Thoracic Paravertebral Block *Prodigy Anesthesia*

Thoracic paravertebral block (TPVB) is a regional anesthetic technique in which local anesthetic is injected in the wedge-shaped space located on either side of the vertebral column close to where the spinal nerves emerge from the vertebral foramen.



TPVB produces both a somatic and sympathetic block on the side of the injection. The spread of the block often extends beyond the dermatome of the spinal nerve and, in some cases, to the contralateral side, lending credence to the belief that some of the local anesthetic spreads through the epidural space as well. The degree of caudal and cephalad spread of a single 15-25 mL injection is highly variable. Because of this, most practitioners inject 3-4 mL at multiple, contiguous thoracic levels.

TPVB is usually performed with the patient in the sitting position for patient comfort and better visualization of anatomical landmarks. The skin is marked about 2.5 cm lateral to the midline of the desired thoracic level. After skin prep and localization of the marked site, a 22-gauge Tuohy or Quincke-tip spinal needle is used for the injection.

The block needle is inserted perpendicular to the skin until it contacts the transverse process of the vertebra one level below the desired thoracic nerve. Because of the risk of missing the transverse process and penetrating the pleura, the needle should not be inserted more than 4 cm before withdrawing the needle and re-trying with a more cephalad or more caudal approach. Once the needle contacts the bone, it should be walked off the process superiorly and advanced until a loss of resistance is encountered as the needle crosses the superior costotransverse ligament. Usually, this is another 1-1.5 cm beyond the transverse process.

Bupivacaine 0.5% or ropivacaine 0.5-0.75% is usually used in volumes of 15-25 mL for a single injection technique or 4-5 mL per level for a multiple-injection technique. This procedure is indicated for postoperative pain control for procedures such as breast surgery, thoracotomy, cholecystectomy, renal surgery, liver surgery, cardiac surgery, neuralgia, and relief of pain due to pleurisy and rib fracture. Absolute contraindications to TPVB include infection at the injection site, local anesthetic allergy, empyema, or neoplasm in the paravertebral space. Coagulopathy and a history of the patient taking anticoagulants are relative contraindications. The procedure carries a higher risk in patients with thoracic deformities such as kyphoscoliosis and in patients who have had previous thoracic surgery. Although intravascular injection, nerve damage, and failed block are all potential complications of a thoracic paravertebral block, the most common is pneumothorax. The aspiration of air during this procedure mandates a chest film to evaluate for pneumothorax.

Hadzic A. Hadzic's Peripheral Nerve Blocks. 2nd ed. New York: McGraw-Hill; 2012: 291-301.

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